Qiyu Wu

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Keywords: NLP; Multilingual; Multimodal; LLMs.

Summary

Research Scientist of Multimodal NLP team at Creative AI Lab, Sony. Ph.D. specializing in NLP, focusing on **Multilingual and Multimodal LLM**. These works have been presented at conferences such as **ACL, EMNLP, NeurIPS, ICLR, NAACL, ICML, TMLR, VLDB, EACL, AAAI,** etc. Served as area chair and reviewer in above top conferences and organized workshops at NeurIPS. Received JSPS DC fellowship.

Research Experiences

Research Scientist, *Sony Group Corporation* 2024.10 – Present

- Leading the architecture and core development of a multimodal content-generation agent, integrating tool orchestration, critique-and-refinement loops, and user-guided revision; collaborating with frontend engineers to deliver an interactive prototype.
- Leading research on multimodal understanding and evaluation with interns and research collaborators, including audio temporal structure modeling and fine-tuning models for generated-content quality assessment. Related work on multimodal understanding, retrieval, and benchmarking: [F10, F9, S9, S8, S7, S6, C11, C10, C7, C6, C5, C4].

Researcher (Part-time), *Institute of Science Tokyo* 2026.11 – Present

- Conducting research on multilingual language model pre-training.

Ph.D. Researcher, *Tsuruoka Lab at UTokyo* 2021.10 – 2024.09

- Word-level preference for LLM-based machine translation [F8]; Word alignment pre-training [F5, S4];
- Text embeddings, generation and compression [S5]; Cross-lingual text embeddings [S2, F6].

Visiting Student, *UC San Diego* 2023.01 – 2023.02

- Generalized diffusion model [C2].

Research Intern, *Microsoft Research Asia* 2020.04 – 2022.02

- Multiple positive examples in sentence embeddings with contrastive learning [F4], work done at STCA.
- Acceleration of language pre-training via better rare word representation [F3], work done at MSR Asia.

R&D Intern, *Baidu Research* 2019.10 – 2020.02

- Multi-task transportation prediction [F2]; Route mode recommendation in Baidu Maps.

Education

The University of Tokyo, Ph.D. Degree in Information Science & Technology. 2021.10 – 2024.09

- Ph.D. at Tsuruoka Lab; JSPS DC2 Fellowship; JST SPRING Fellowship.

Peking University, M.Sc. Degree in Computer Software & Theory. 2018.09 – 2021.07

Sichuan University, B.Eng. Degree in Computer Science & Technology. 2014.09 – 2018.07

Advising and Services

Advised Students: Youngtaek Oh [S7], Daeyong Kwon [S6], Dominick Reilly [S9], Jongsuk Kim [S8], Zhongtao Miao [S5, S4, S2], Kaiyan Zhao [F6, S3], Yutao Xie [S1].

Area Chair of ACL Rolling Review (2025 – Present); NeurIPS (2026 – Present).

Session Chair at EMNLP 2026, "Semantics: Lexical and Sentence-Level". 2026

NeurIPS 2026 Top Area Chair. 2026

ACL 2026 Outstanding Reviewer (Top 1.4%). 2026

Reviewer of ACL Rolling Review, EMNLP, ICLR, NeurIPS, ICML, etc. 2021 – Present

Organizer of "NeurIPS 2025 GenProCC Workshop". 2025

DC2 Fellowship of JSPS. 2024

Teaching Assistant of "Social Implications of Computing" at Peking University. 2019

Publications

‡ **indicates corresponding author;** ∗ **indicates equal contribution.**

(Co-)First

[F10] **Qiyu Wu**, Shuyang Cui, Satoshi Hayakawa, Wei-Yao Wang, Hiromi Wakaki, Yuki Mitsufuji. "MCA: Modality Composition Awareness for Robust Composed Multimodal Retrieval", ***(EMNLP 2026 Main, Oral)***.

[F9] **Qiyu Wu**, Mengjie Zhao, Yutong He, Lang Huang, Junya Ono, Hiromi Wakaki, Yuki Mitsufuji. "Towards Reporting Bias in Visual-language Datasets: Bi-modal Data Augmentation by Decoupling Object-attribute Association", ***(ICCV 2025 Workshop Multimodal Representation and Retrieval)***.

[F8] **Qiyu Wu**, Masaaki Nagata, Zhongtao Miao, Yoshimasa Tsuruoka. "Word Alignment as Preference for Machine Translation", ***(EMNLP 2024 Main, Oral)***.

[F7] **Qiyu Wu**, Zilong Wu, Yoshimasa Tsuruoka. "SGA: Scene Graph Alignment for Evaluation of Text-to-Image Generation", ***(KDD 2024 Workshop HI-AI)***.

[F6] Kaiyan Zhao∗, **Qiyu Wu**∗, Xin-Qiang Cai, Yoshimasa Tsuruoka. "Leveraging Multi-lingual Positive Instances in Contrastive Learning to Improve Sentence Embedding", ***(EACL 2024 Main)***.

[F5] **Qiyu Wu**, Masaaki Nagata, Yoshimasa Tsuruoka. "WSPAlign: Word Alignment Pre-training via Large-Scale Weakly Supervised Span Prediction", ***(ACL 2023 Main)***.

[F4] **Qiyu Wu**, Chongyang Tao, Tao Shen, Can Xu, Xiubo Geng, Daxin Jiang. "PCL: Peer-Contrastive Learning with Diverse Augmentations for Unsupervised Sentence Embeddings", ***(EMNLP 2022 Main)***.

[F3] **Qiyu Wu**∗, Chen Xing∗, Yatao Li, Guolin Ke, Di He, Tie-Yan Liu. "Taking Notes on the Fly Helps Language Pre-Training", ***(ICLR 2021)***.

[F2] Hao Liu∗, **Qiyu Wu**∗, Fuzhen Zhuang, Xinjiang Lu, Dejing Dou, Hui Xiong. "Community-Aware Multi-Task Transportation Demand Prediction", ***(AAAI 2021)***.

[F1] Dechuan Yang∗, **Qiyu Wu**∗, Wei Chen, Tengjiao Wang, Zhen Qiu, Di Liu, Yingbao Cui. "Stance Detection with Stance-wise Convolution Network", ***(NLPCC 2020)***.

Advised Research

[S9] Dominick Reilly, **Qiyu Wu**‡, Hiromi Wakaki, Srijan Das, Yuki Mitsufuji. "Mixture of Probes: Learning from Privileged Modalities in Multimodal LLMs Through Probing", ***(NeurIPS 2026)***.

[S8] Jongsuk Kim, **Qiyu Wu**‡, Zhuoyuan Mao, Hiromi Wakaki, Junmo Kim, Yuki Mitsufuji. "MLLMCLIP: Feature-Level Distillation of MLLM for Robust Vision-Language Representations", ***(EMNLP 2026 Main)***.

[S7] Youngtaek Oh, **Qiyu Wu**‡, Hiromi Wakaki, Junmo Kim, Yuki Mitsufuji. "Syn-Omni: Structured Specialization and Progressive Collaboration for Omnimodal Embeddings", ***(EMNLP 2026 Findings)***.

[S6] Daeyong Kwon, **Qiyu Wu**‡, Shinobu Kuriya, Junghyun Koo, Shuyang Cui, Zhi Zhong, Wei-Hsiang Liao, Hiromi Wakaki, Yuki Mitsufuji. "MusTBench: Benchmarking and Advancing Temporal Grounding in Music LLMs", ***manuscript (2026)***.

[S5] Zhongtao Miao, **Qiyu Wu**, Yoshimasa Tsuruoka. "GRC: Unifying Reasoning-Driven Generation, Retrieval and Compression", ***manuscript (2026)***.

[S4] Zhongtao Miao, **Qiyu Wu**, Masaaki Nagata, and Yoshimasa Tsuruoka. "Improving Word Alignment Using Semi-Supervised Learning", ***(ACL 2025 Findings)***.

[S3] Kaiyan Zhao, **Qiyu Wu**, Zhongtao Miao, Yoshimasa Tsuruoka. "Prompt Tuning Can Simply Adapt Large Language Models to Text Encoders", ***(RepL4NLP 2025)***.

[S2] Zhongtao Miao, **Qiyu Wu**, Kaiyan Zhao, Zilong Wu, Yoshimasa Tsuruoka. "Enhancing Cross-lingual Sentence Embedding for Low-resource Languages with Word Alignment", ***(NAACL 2024 Findings)***.

[S1] Yutao Xie, **Qiyu Wu**, Wei Chen and Tengjiao Wang. "Stable Contrastive Learning for Self-Supervised Sentence Embeddings With Pseudo-Siamese Mutual Learning", ***(TASLP 2022)***.

Collaborative

[C11] Ce Zhang, Ziyang Wang, Yulu Pan, Oluwatumininu Oguntola, Pranav Wagh, **Qiyu Wu**, Hiromi Wakaki, Mohit Bansal, Gedas Bertasius. "Searching Videos as Trees: Self-Correcting Agents for Grounded Long Video QA", ***(NeurIPS 2026)***.

[C10] Silin Gao, Hao Zhao, Zeming Chen, Sepideh Mamooler, Antara Raaghavi Bhattacharya, **Qiyu Wu**, Hiromi Wakaki, Yuki Mitsufuji, Li Mi, Syrielle Montariol, Antoine Bosselut. "DynaVieW: Schema-Guided World Modeling for Understanding Hierarchical Visual Dynamics", ***(ICML 2026).***

[C9] Shaomu Tan, Ryosuke Mitani, Ritvik Choudhary, **Qiyu Wu**, Toshiyuki Sekiya, Christof Monz. "Remedy-R: Generative Reasoning for Machine Translation Evaluation without Error Annotations", ***(ACL 2026).***

[C8] Shuyang Cui, Zhi Zhong, **Qiyu Wu**, Zachary Novack, et al. "BREAK-THE-BEAT! Controllable MIDI-to-Drum Audio Synthesis", ***(ICASSP 2026).***

[C7] Seungheon Doh, Junghyun Koo, Marco A Martínez-Ramírez, Woosung Choi, Wei-Hsiang Liao, **Qiyu Wu**, Juhan Nam, Yuki Mitsufuji. "LLM2Fx-Tools: Tool Calling For Music Post-Production", ***(ICLR 2026)***.

[C6] Wei-Yao Wang, Kazuya Tateishi, **Qiyu Wu**, Shusuke Takahashi, Yuki Mitsufuji. "VIRTUE: Visual-Interactive Text-Image Universal Embedder", ***(ICLR 2026)***.

[C5] Zhuoyuan Mao, Mengjie Zhao, **Qiyu Wu**, Hiromi Wakaki, Yuki Mitsufuji. "DeepResonance: Enhancing Multimodal Music Understanding via Music-centric Multi-way Instruction Tuning", ***(EMNLP 2025 Main)***.

[C4] Zhuoyuan Mao, Mengjie Zhao, **Qiyu Wu**, Zhi Zhong, Wei-Hsiang Liao, Hiromi Wakaki, Yuki Mitsufuji. "Cross-Modal Learning for Music-to-Music-Video Description Generation", ***(RepL4NLP 2025)***.

[C3] Yuting Tang, Xin-Qiang Cai, Yao-Xiang Ding, **Qiyu Wu**, Guoqing Liu, Masashi Sugiyama. "Reinforcement Learning from Bagged Reward", in Transactions on Machine Learning Research ***(TMLR 2025; ICML 2026 Poster Track)***.

[C2] Guangyi Liu, Yu Wang, Zeyu Feng, **Qiyu Wu**, Liping Tang, Yuan Gao, Zhen Li, Shuguang Cui, Julian McAuley, Eric P. Xing, Zichao Yang, Zhiting Hu. "Unified Generation, Reconstruction, and Representation: Generalized Diffusion with Adaptive Latent Encoding-Decoding", ***(ICML 2024)***.

[C1] Shiwen Wu, **Qiyu Wu**, Honghua Dong, Wen Hua, Xiaofang Zhou. "Blocker and Matcher Can Mutually Benefit: A Co-Learning Framework for Low-Resource Entity Resolution.", ***(VLDB 2024 Research Track)***.

F: (Co-)First; S: Advised research; C: Collaborative.